

EX.NO: 6

IMPLEMENTATION OF DECISION TREE CLASSIFICATION ALGORITHM

AIM:

To classify the new instance (new sample) to a target data (grape fruit or orange) using decision tree algorithm in R Studio.

PROCEDURE:

1. Install the necessary package in R Studio such as “rpart” and “rpart.plot”
2. Set the current working directory
3. Save the dataset in the current working directory
4. Read the dataset in the form of csv file using **read.csv ()**
5. Finally create the decision tree and predict the category for the new instance.

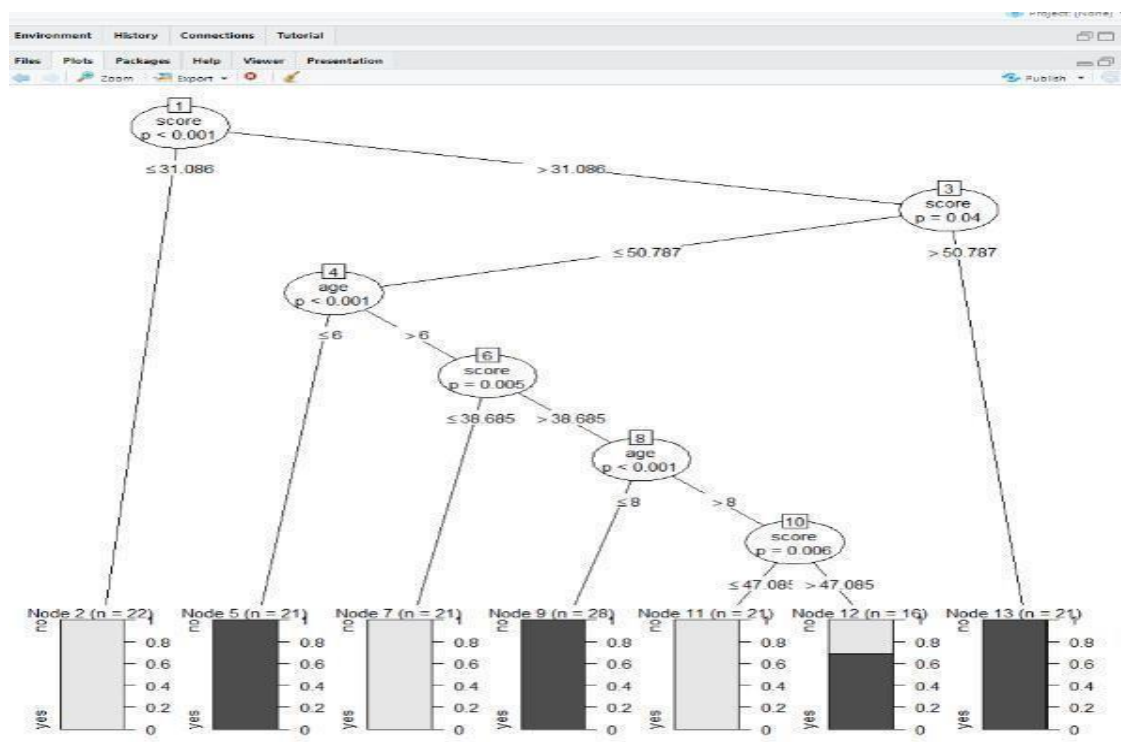
SOURCE CODE:

```
library(datasets)
library(caTools)
library(party)
library(dplyr)
library(magrittr)
data("readingSkills")
```

```
head(readingSkills)
sample_data = sample.split(readingSkills, SplitRatio =
0.8) train_data <- subset(readingSkills, sample_data ==
TRUE) test_data <- subset(readingSkills, sample_data
== FALSE) model<- ctree(nativeSpeaker ~ ., train_data)
plot(model)
ctree(formula,data
)
```

OUTPUT:

```
warning in install.packages :  
package 'closure' is not available for this version of R  
  
A version of this package for your version of R might be available elsewhere,  
see the ideas at  
https://cran.r-project.org/doc/manuals/r-patched/R-admin.html#installing-packages  
> library(datasets)  
> library(caTools)  
> library(party)  
> library(dplyr)  
> library(magrittr)  
>  
> data("readingskills")  
> head(readingskills)  
  nativespeaker age shoeSize score  
1          yes   5  24.83189 32.29385  
2          yes   6  25.95238 36.63105  
3          no  11  30.42170 49.60593  
4          yes   7  28.66450 40.28456  
5          yes  11  31.88207 55.46085  
6          yes  10  30.07843 52.83124  
> sample_data = sample.split(readingskills, splitRatio = 0.8)  
> train_data <- subset(readingskills, sample_data == TRUE)  
> test_data <- subset(readingskills, sample_data == FALSE)  
> model <- ctree(nativespeaker ~ ., train_data)  
> plot(model)  
> ctree(formula, data)  
Error in ctree(formula, data) : object of type 'closure' is not subtable
```



RESULT:

Thus the new instance has been classified to a target attribute (grape fruit) using Decision Tree algorithm.